

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

MARK SCHEME

CLASS

TUTOR'S
NAME

CHEMISTRY

8872/02

Paper 2 Structured Questions

12 September 2011

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet.
Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Section A	
1	9
2	11
3	5
4	4
5	5
6	6
Total	40

This document consists of **13** printed pages and **1** blank page

[Turn over

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 Elements **A**, **B**, **C**, **D** and **E** have consecutive atomic numbers below 20. The table below shows the first four ionisation energies of elements **A**, **B**, **C**, **D** and **E**.

Element	First I.E.	Second I.E.	Third I.E.	Fourth I.E.
A	736	1450	7740	10500
B	577	1820	2740	11600
C	786	1580	3230	4360
D	1060	1900	2920	4960
E		2260	3390	4540

- (a) Identify the group to which element **A** belongs. Explain your reasoning.

Group: **II** **[1]**

Explanation:

The largest difference is between the 2nd and 3rd IEs. The 3rd electron must have been removed from the inner quantum shell. Therefore there are 2 valence electrons, hence **A** is in group II. **[1]**

[2]

- (b) Suggest a possible value for the missing first ionisation energy of **E**. Explain your answer.

1st I.E. of **E**: Accept **E** 800-1000. Actual value: 10000. (not too low because it's just a small "dip") **[1]**

D: $ns^2np_x^1np_y^1np_z^1$

E: $ns^2np_x^2np_y^1np_z^1$ **[1]**

Less energy is required to remove the paired 2p_x electron in **E** as it experiences inter-electronic repulsion. **[1]**

[3]

- (c) Explain the difference in **second** ionisation energy between elements **A** and **B**.

For second I.E., electrons are removed from the X⁺ ion.

A⁺: ns^1

B⁺: ns^2

B⁺ has a higher nuclear charge and screening effect remains approximately constant as valence electrons are in the same outershell **[1]**, hence net attraction of outermost electrons to nucleus increases, more energy is required to remove electron in **B**⁺. Hence 2nd I.E. of **B**⁺ is higher than **A**⁺. **[1]**

[2]

- (d) Either of the oxides of **A** or **E** is soluble in aqueous sodium hydroxide. Identify this oxide and write an equation to explain why it dissolves.

Oxide of E is soluble in NaOH(aq). [1]

E is in group VI. It's an acidic (non-metal) oxide that dissolves in NaOH.
It may form EO₂ or EO₃.

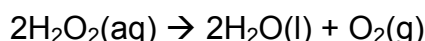
Equation: EO₂ (s) + 2 NaOH (aq) → NaEO₃ (aq) + H₂O (l)

OR EO₃ (s) + 2 NaOH (aq) → NaEO₄ (aq) + H₂O (l) [1]

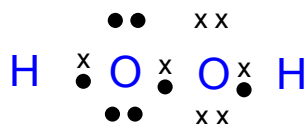
[2]

[Total: 9]

- 2 A dilute solution of hydrogen peroxide can be used to bleach hair. It decomposes slowly in aqueous solution according to the following equation.



- (a) Draw the dot-and-cross diagram to illustrate the bonding in a H_2O_2 molecule, showing the outermost shell electrons only.



[1]

- (b) A solution with an original concentration of 3.0 mol dm^{-3} was placed in a bottle contaminated with transition metal ions, which act as catalyst for the decomposition. The rate of decomposition was measured by withdrawing 10 cm^3 portions at various times and titrating with acidified $0.100 \text{ mol dm}^{-3} \text{KMnO}_4(\text{aq})$.

- (i) Write the balanced equation for the reaction between MnO_4^- and H_2O_2 .



- (ii) Calculate the concentration of H_2O_2 when 30.0 cm^3 of $0.100 \text{ mol dm}^{-3} \text{KMnO}_4$ reacts with 10.0 cm^3 of H_2O_2 .

$$\frac{c \times 10}{0.1 \times 30} = \frac{5}{2}$$

$$c = 0.75 \text{ mol dm}^{-3} \quad [1]$$

- (iii) Given that the half-life for the decomposition is 12 minutes, estimate how long the solution had been in the contaminated bottle.

$$3.0 \rightarrow 1.5 \rightarrow 0.75$$

2 half-life has passed. [1]

Hence the solution has been in $2(12) = 24$ minutes in the contaminated bottle. [1]

[4]

- (c) Explain, with reference to the *Data Booklet*, why the decomposition of H_2O_2 is an exothermic process.

BE of $(\text{O}=\text{O}) = +496 \text{ kJ mol}^{-1}$

BE of $(\text{O}-\text{O}) = +150 \text{ kJ mol}^{-1} \quad [1]$

Bonds formed $(\text{O}=\text{O})$ is stronger than bonds broken $(\text{O}-\text{O}) \quad [1]$

OR

$$\Delta H_r = \sum \text{Bonds Broken} - \sum \text{Bonds Formed}$$

$$= [4(460) + 2(150)] - [4(460) + 496] \quad [1]: \text{correct BE values}$$

$$= -196 \text{ kJ mol}^{-1}$$

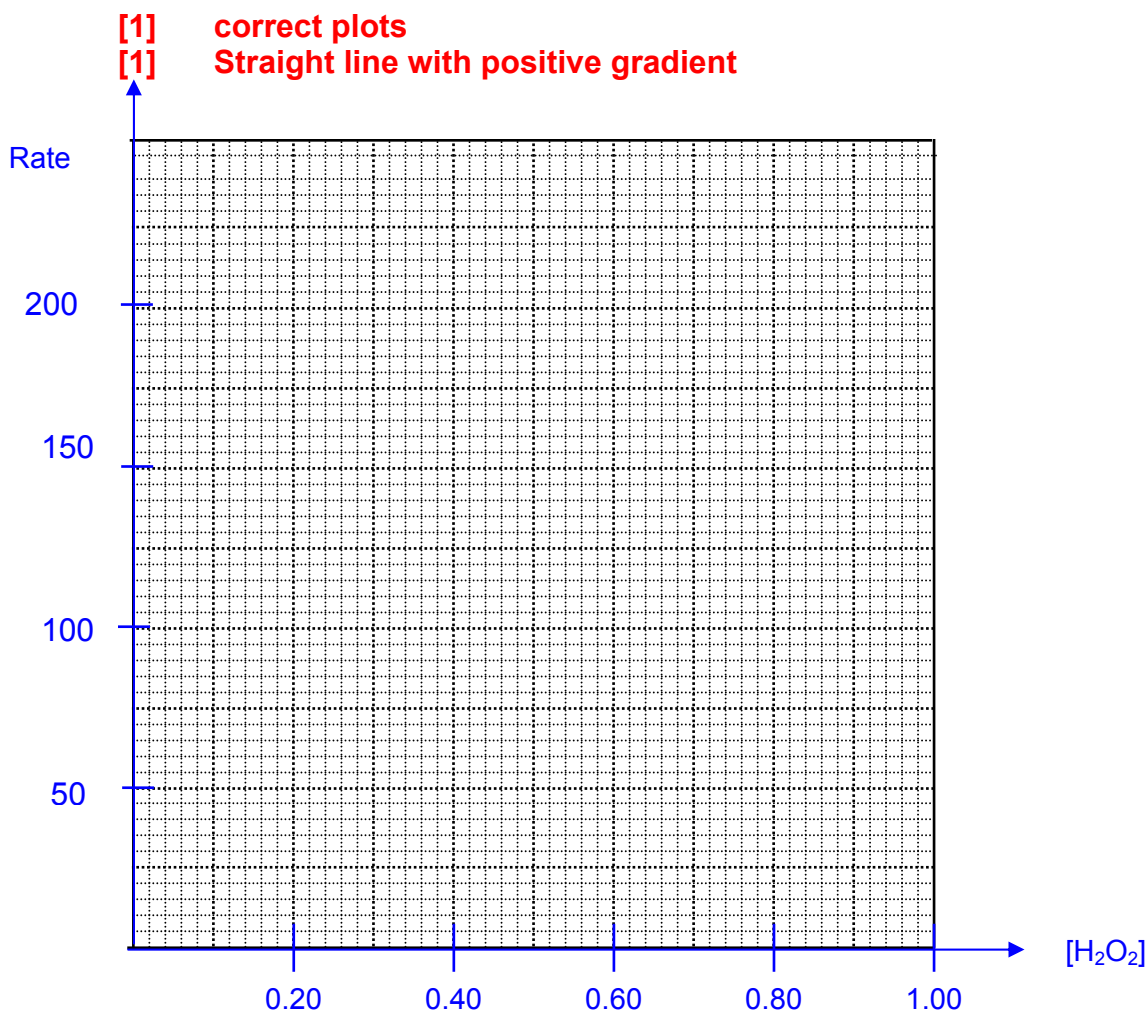
[1] Correct enthalpy change value

[2]

- (d) In order to determine the rate equation for this reaction, an investigation was carried out at constant temperature. The following results were obtained.

Initial rates/ $\text{mol dm}^{-3} \text{ s}^{-1}$	16.2	47.2	78.8	124	172
$[\text{H}_2\text{O}_2]/$ mol dm^{-3}	0.09	0.25	0.44	0.67	0.90

- (i) To determine the order of reaction with respect to $[\text{H}_2\text{O}_2]$, use these data to plot a suitable graph on the grid below.



[2]

- (ii) What is the order of reaction with respect to $[\text{H}_2\text{O}_2]$?
Explain your answer.

$$\text{Rate} = k [\text{H}_2\text{O}_2]$$

Straight line passing through origin for rate against $[\text{H}_2\text{O}_2]$

Hence rate $\propto [\text{H}_2\text{O}_2]$

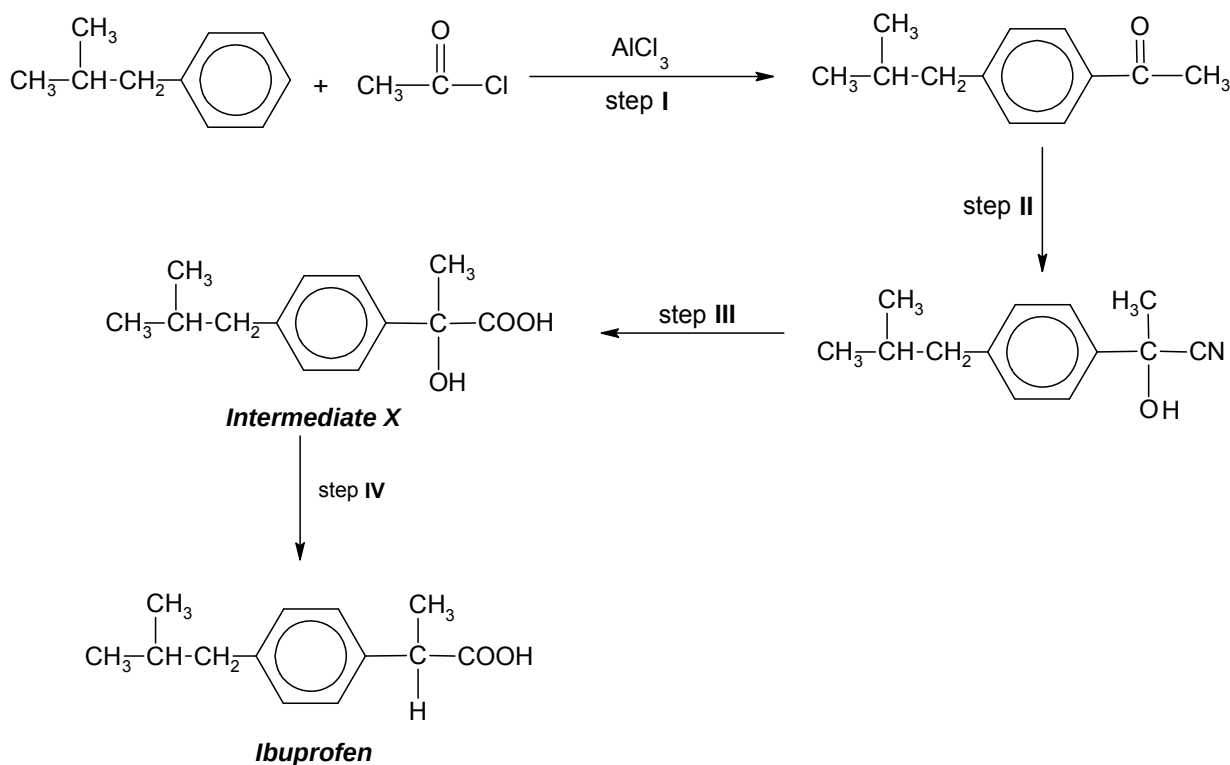
[1] correct rate equation written

[1] establish direct proportional

[2]

[Total:11]

- 3 Ibuprofen is an anti-inflammatory drug used to treat fever and arthritis. The following scheme shows the synthesis of Ibuprofen from iso-butylbenzene.



- (a) Suggest the type of reaction which occurs in step I and IV.

Step I: Substitution

Step IV: Reduction

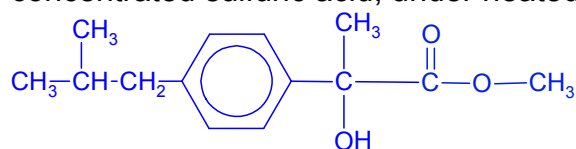
[2]

- (b) Suggest the reagents and conditions needed for step II and III.

Step	Reagent and condition
II	HCN, trace amount of NaOH/NaCN, 10-20°C
III	H ₂ SO ₄ (aq), heat under reflux

[2]

- (c) Draw the structural formula of the product formed when 1 mole of intermediate X (product of step III) reacts with 1 mole of methanol in the presence of concentrated sulfuric acid, under heated conditions.

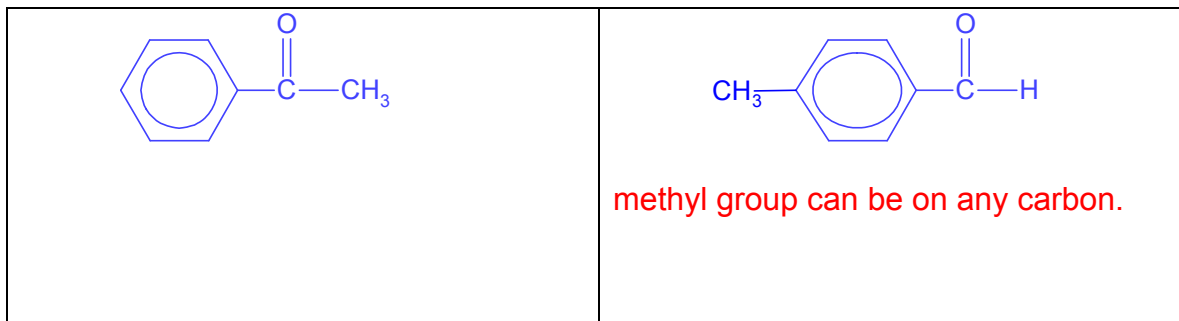


[1]

[Total: 5]

- 5 (a) An aromatic organic compound **K** has the molecular formulae C_8H_8O . **K** gives a orange crystalline solid with 2,4-Dinitrophenylhydrazine but does **not** react with Fehling's solution.

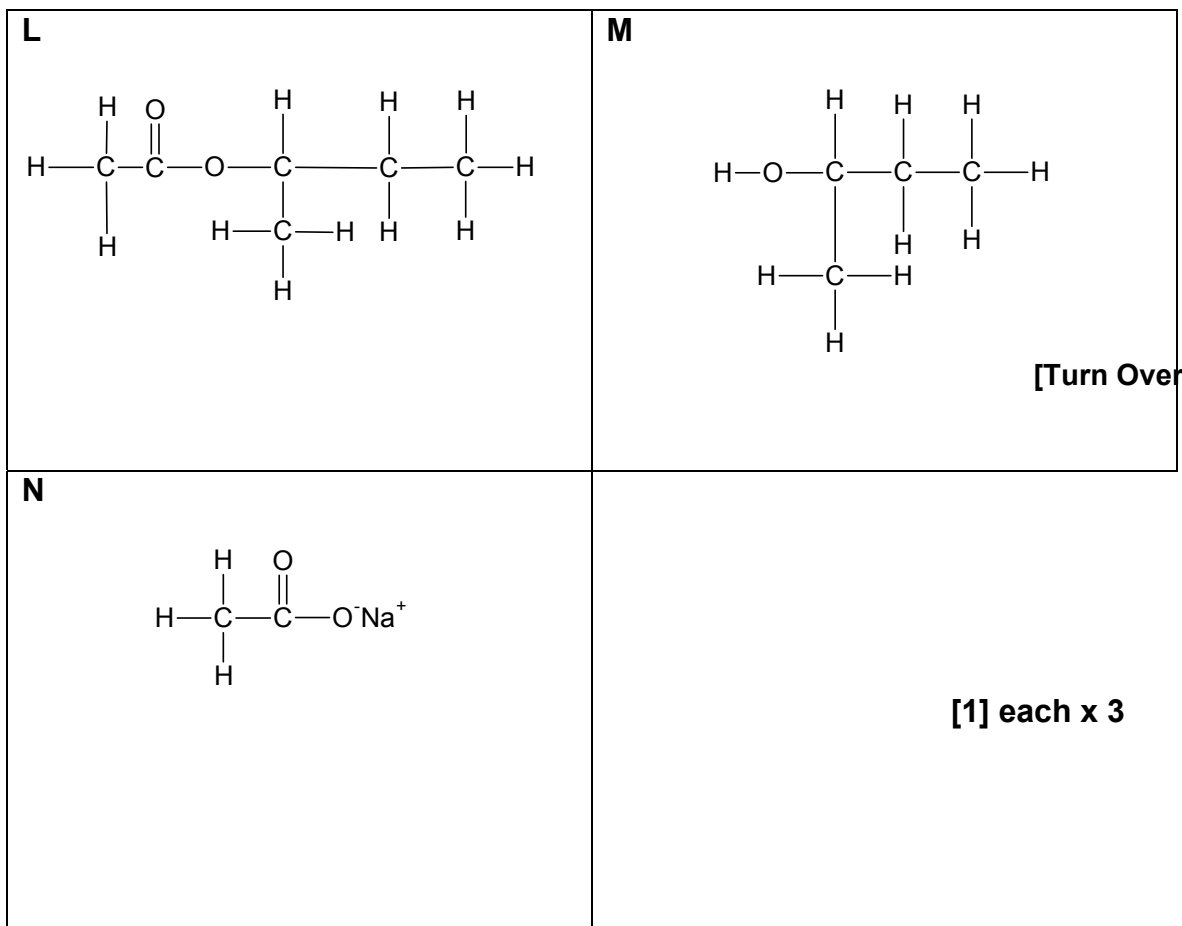
Suggest 2 possible identities of **K**. The 2 structures should contain 2 different functional groups.



[2]

- (b) Another organic compound **L**, $C_6H_{12}O_2$, reacts with hot aqueous sodium hydroxide to form 2 products **M** and **N**. **M** has the molecular formulae $C_4H_{10}O$ and it produces a yellow crystalline solid with aqueous alkaline iodine.

Identify **L**, **M** and **N** and draw the displayed formula.

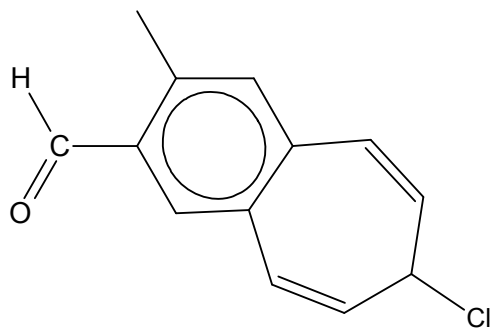


[3]

[Total: 5]

[Turn Over]

- 6(a) In the spaces provided below, draw the organic products formed when compound **J** is reacted with each of the following reagents.

Compound **J**

(i) Heat with ammonia dissolved in ethanol in a sealed tube [1] substitution	
(ii) Acidified potassium manganate(VII) [1] 2 side chain oxidation [1] alkene oxidation [1] oxidised segment	

[4]

- (b) A white precipitate is observed when Compound **A** is heated under reflux with ethanolic silver nitrate.

State and explain the rate of this reaction changes when the chloro group in compound **A** is replaced by a bromo group.

Rate of hydrolysis increases. **[1]**

C—Br bond is weaker than C—Cl bond. Hence it is easier to break the bond, hence faster rate of hydrolysis. **[1]**

[2]
[Total:6]

Section B

Answer **two** questions from this section on separate answer paper.

- 7 a) Define the term *mole*.
 Why is the phrase “the mass of one mole of oxygen” ambiguous? [3]
- One mole of a substance is the amount of substance that contains the same number of particles as there are atoms in 12 g of the carbon-12 isotope. [1]
 The phrase is ambiguous because the chemical formula of the substance is not specified [1];
 “one mole of oxygen” could be referring oxygen atoms, O, or oxygen molecules, O₂. [1]
- (b) By using the data in this question, estimate how much oxygen a person requires on average in a life-time and how far a motor car can go using the same amount of oxygen.
 The typical daily food requirement of a person can be considered to be 1.2 kg of carbohydrate. The person obtains energy by the oxidation of the carbohydrate, which can be represented by the formula (CH₂O)_n.
- (i) Construct an equation for the complete oxidation of the carbohydrate, (CH₂O)_n.

$$(\text{CH}_2\text{O})_n + n\text{O}_2 \rightarrow n\text{CO}_2 + n\text{H}_2\text{O}$$
 [1]
 *No marks if not balanced with “n”
- (ii) Given that the *empirical relative formula mass* of the carbohydrate is 30, calculate the number of moles of oxygen required by the person each day.

$$M_r \text{ of carbohydrate} = 30n$$

$$\text{Amount of carbohydrate} = \frac{1200}{30n} = \frac{40}{n} \text{ mol}$$

$$\text{Amount of O}_2 = \frac{40}{n} \times n = 40 \text{ mol}$$
 [1]
 *No marks if M_r of carbohydrate did not indicate “n”
- (iii) Hence determine how many moles of oxygen a person require in a life-time of 70 years? [3]

$$\text{Amount of O}_2 = 40 \times 70 \times 365 = 1.022 \times 10^6 \text{ mol}$$
 [1]
- (c) A typical motor car requires 6 dm³ of octane, C₈H₁₈, to travel 100 km.
 (i) Construct an equation for the complete combustion of octane.

$$\text{C}_8\text{H}_{18} + \frac{25}{2} \text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$$
 [1]

- (ii) The density of octane is 0.7 g cm^{-3} .

Calculate how many moles of oxygen would be required for a typical car using octane to travel 100 km.

$$\text{Mass of octane to travel 100 km} = 0.7 \times 6 \times 1000 = 4200\text{g}$$

$$\text{Amount of octane} = \frac{4200}{114} = 36.84 \text{ mol [1]}$$

$$\text{Amount of O}_2 \text{ required} = 36.84 \times \frac{25}{2} = 460.5 \text{ mol [1]}$$

- (iii) From your answer in b(iii), calculate how many kilometers the car can travel using the same amount of oxygen a person uses in a life-time.

[4]

Since 460.5 mol of O_2 needed to run 100 km, therefore using

$$1.022 \times 10^6 \text{ mol of O}_2 \text{ can run} = \frac{1.022 \times 10^6}{460.5} \times 100 = 2.22 \times 10^5 \text{ km [1]}$$

- (d) Due to worldwide decline in petroleum production, scientists are trying to come up with more energy efficient octane fuel.

In one experiment, 10 cm^3 of a hydrocarbon C_8H_y , were allowed to react with an excess of oxygen at 150°C and 1 atm. There was an expansion of 30 cm^3 . Deduce the value of y .

[4]



Let volume of unreacted oxygen be $q \text{ cm}^3$.

Since volume of $\text{C}_8\text{H}_y = 10 \text{ cm}^3$,

Volume of reacted $\text{O}_2 = 80 + \frac{10y}{4} \text{ cm}^3$,

Volume of CO_2 produced = 80 cm^3 ,

Volume of H_2O produced = $5y \text{ cm}^3$.

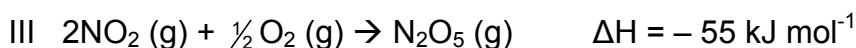
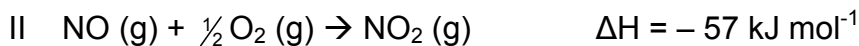
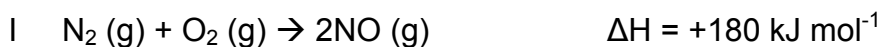
[1] for correcting deducing the volume of all the species, O_2 , CO_2 and H_2O

Final volume – initial volume = 30 cm^3 ,

$$(q + 80 + 5y) - (10 + q + 80 + \frac{10y}{4}) = 30 \text{ [1] for equation}$$

$$y = 16 \text{ [1]}$$

- (e) Dinitrogen pentoxide, N_2O_5 , can be produced by the following reaction sequence.



- (i) Explain why reaction I occurs readily in car engines, but not in the atmosphere.

Due to the high temperature of the car engine [1]

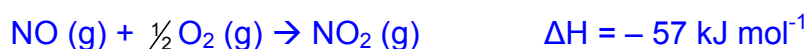
sufficient energy is provided for breaking of the strong N≡N and O=O bonds. [1]

This energy is not readily available in the atmosphere.

- (ii) What is meant by *the standard enthalpy change of formation* of a compound?

The energy change when one mole of a compound is formed from its constituent elements in their standard states under standard conditions. [1]

- (iii) Use the data given to calculate the enthalpy change of formation of nitrogen dioxide.



$$\Delta H_r = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

$$-57 = \Delta H_f(\text{NO}_2) - [\Delta H_f(\text{NO}) + \frac{1}{2} \Delta H_f(\text{O}_2)] \quad \text{[1] correct substitution}$$

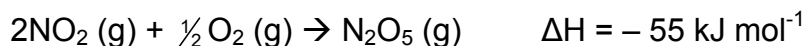
$$-57 = \Delta H_f(\text{NO}_2) - \left[\frac{180}{2} + \frac{1}{2}(0) \right]$$

$$\Delta H_f(\text{NO}_2) = -57 + 90 = +33 \text{ kJ mol}^{-1}$$

[1] for correct calculation with units

- (iv) Hence, calculate the enthalpy change of formation of dinitrogen pentoxide,

[6]



$$\Delta H_r = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

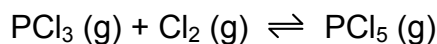
$$-55 = \Delta H_f(\text{N}_2\text{O}_5) - [2\Delta H_f(\text{NO}_2) + \Delta H_f(\text{O}_2)] \quad \text{[1] correct substitution}$$

$$-55 = \Delta H_f(\text{N}_2\text{O}_5) - [2(33) + 0]$$

$$\Delta H_f(\text{N}_2\text{O}_5) = -55 + 2(33) = +11 \text{ kJ mol}^{-1}$$

[1] for correct calculation with units

- 8 (a) Phosphorous pentachloride is one of the most important phosphorus chlorides as it is used widely as a chlorinating agent. When PCl_3 and Cl_2 was placed in a 4 dm^3 reaction vessel and heated to 250°C , it was found that $1.68 \times 10^{-4} \text{ mol}$ of PCl_5 , 0.0520 mol of PCl_3 and 0.0156 mol of Cl_2 was present in the following equilibrium.



- (i) Write the K_c expression for this equilibrium.

$$K_c = \frac{[\text{PCl}_5]}{[\text{PCl}_3][\text{Cl}_2]} \quad [1]$$

- (ii) Calculate the K_c value at 250°C and state the units.

$$[1]+[1] \quad K_c = \frac{\left(\frac{1.68 \times 10^{-4}}{4}\right)}{\left(\frac{0.0520}{4}\right)\left(\frac{0.0156}{4}\right)} = 1.21 \text{ mol}^{-1} \text{ dm}^3$$

- (iii) Explain, using Le Chatelier's Principle, how the position of equilibrium and K_c value changes when the size of the reaction vessel is increased to 10 dm^3 .

By Le Chatelier's Principle, an increase in volume would favour the reaction that produces greater amounts of gaseous particles, [1]

Thus the backward reaction is favoured and POE shifts to the left. [1]

More reactants are present than products thus K_c value decreases. [1]

- (iv) When the temperature decreases, more PCl_5 was formed. Suggest with reason if the forward reaction is exothermic or endothermic. [8]

$$\Delta H = -123.8 \text{ kJ mol}^{-1}$$

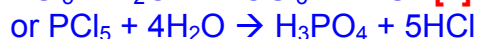
By Le Chatelier's Principle, a decrease in temperature would favour the exothermic reaction. [1]

Since the forward reaction is favoured in producing more PCl_5 , the forward reaction is exothermic. [1]

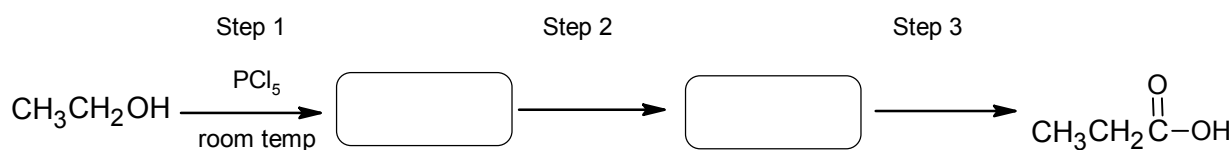
- (b) PCl_5 reacts readily in water to produce 2 mol of HCl which dissolved in water to give a solution of $\text{pH} = 1.7$.

- (i) Name the reaction and write a balanced equation for the reaction.

Hydrolysis. [1]



- (ii) PCl_5 is an important reactant in the production of propanoic acid, $\text{C}_3\text{H}_6\text{O}_2$, from ethanol as seen from the scheme below:



Name the type of reaction in Step 1.

Substitution [1]

- (iii) Give the structures of the product for Steps 1 and 2.

Step 1: $\text{CH}_3\text{CH}_2\text{Cl}$ [1]

Step 2: $\text{CH}_3\text{CH}_2\text{CN}$ [1]

- (iv) State the reagents and conditions for Steps 2 to 3 [7]

Step 2: NaCN in ethanol, heat [1]

Step 3: H_2SO_4 or HCl , heat [1]

- (c) Organic acids are usually weak acids. A 25 cm^3 sample of propanoic acid produced in part (b) was found to weigh 0.0370 g and has a pH value of 3.5 .

- (i) Show, with explanation, that propanoic acid is also a weak acid.

$$[\text{Propanoic acid}] = \frac{0.0370}{\frac{74}{25} \times \frac{1}{1000}} = 0.0200 \text{ mol dm}^{-3} \quad [1/2]$$

$$[\text{H}^+] = 10^{-3.5} = 3.16 \times 10^{-4} \text{ mol dm}^{-3} \quad [1/2]$$

Since $[\text{Propanoic acid}] \gg [\text{H}^+]$ [1/2]

not all the molecular propanoic acid dissociated to give H^+ [1/2],
propanoic acid is a weak acid.

- (ii) When half the amount of the propanoic acid reacted with equal amount of sodium hydroxide, it was found that the resulting solution pH did not change much when a small amount of acid or alkali was added to it.

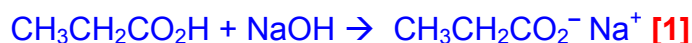
With the aid of equations, explain why the above observation occurs.

[5]

The small amount of acid added to the solution would react with the large reservoir of $\text{CH}_3\text{CH}_2\text{CO}_2^- \text{Na}^+$ to give a neutral solution, thus keeping the pH of the solution almost constant. [$\frac{1}{2}$]



The small amount of alkali added to the solution would react with the large reservoir of propanoic acid to give a neutral salt and water, thus keeping the pH of the solution almost constant. [$\frac{1}{2}$]



- 9 (a) Some physical properties of the oxides of elements **M**, **Q**, **X** and **Y** are given below.

	Formula of oxide	Melting point / °C	Appearance at room temperature	Conductivity in the molten state
(i)	MO₂	1700	White solid	None
(ii)	Q₂O	0	Colourless liquid	None
(iii)	XO	2850	White solid	Good
(iv)	YO₂	−73	Colourless gas	None

For each of the following of these four oxides, what are their structures, bonding and particles (ions, atoms or molecules) present?

[4]

	Formula of oxide	Structure	Bonding	Particles
(i)	MO ₂	Macromolecular	Covalent between	M and O atoms
(ii)	Q ₂ O	Molecular	Intramolecular covalent and <u>hydrogen</u> bonding between	Q ₂ O molecules
(iii)	XO	(giant) Ionic	Ionic bonds between	The cations X ²⁺ and anions O ^{2−}
(iv)	YO ₂	Molecular	Intramolecular covalent and van der Waal bonding between	YO ₂ molecules

- (b) For the reaction $2\text{HI (g)} \rightleftharpoons \text{H}_2\text{(g)} + \text{I}_2\text{(g)}$

- (i) The enthalpy change is +9.6 kJ mol^{−1}, the activation energy for the forward reaction is +182.8 kJ mol^{−1}. Calculate the activation energy of the backward reaction.

$$(+182.8) - (+9.6) = +173.2 \text{ kJ mol}^{-1}$$

- (ii) In the presence of a gold catalyst, the activation energy for the backward reaction is +85.3 kJ mol^{−1}. Calculate the activation energy of the forward reaction of the catalysed reaction.

$$(+85.3) + (+9.6) = +94.9 \text{ kJ mol}^{-1}$$

- (iii) Sketch the energy profile for the reaction, label all the values given and calculated in (i) and (ii).

1 m for correct graphs of uncatalysed and catalysed reaction
1 m for all values correctly entered.

- (iv) Explain, with the aid of a sketch of the Boltzmann distribution, how the presence of the gold catalyst increase the rate of the decomposition.

[8]

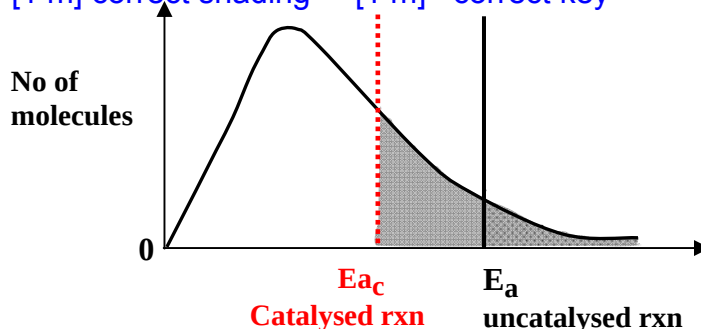
Solution

- In the presence of a catalyst, the reaction has a lower activation energy compared to the uncatalysed reaction.
- **[1]** More molecules will possess energy greater than or equal to this lowered activation energy,
- **[1]** hence frequency of activated collisions will increase.

Boltzmann Distribution Diagrams

no marks at all for incorrect/ unlabelled axes

[1 m] correct shading [1 m] correct key



Key:

 No of molecules with energy $\geq E_a$

 No of molecules with energy $\geq E_{a_c}$ (catalysed)

- (b) “Cracking” in the petroleum industry refers to reactions which convert abundant long chain molecules, which have limited uses’ to short chain, economically useful fractions. Commonly long chain alkanes are broken up into a mixture that could consist of smaller alkanes, alkenes and or hydrogen gas.

The hydrocarbon, $C_{10}H_{22}$ was subjected to cracking and 3 non-cyclic products were obtained.

A had a mass of 58 and did not decolourise bromine in carbon tetrachloride.

B had a mass of 56 and decolourised bromine in carbon tetrachloride.

- (i) Give a possible structure for **A**.
any non-cyclic C_4H_{10} structure.

- (ii) Give 2 possible structures for **B**.

any 2 non-cyclic C_4H_8 isomers, containing 1 C=C bond.

- (iii) Using only molecular formula write a balanced equation for the cracking process.

[4]



- (d) Propan-2-ol is found in nail polish while propane-1,2-diol is a component used in softening lotion. These two alcohols are synthesized from propene.

- (i) State the reagent and conditions for the formation of propan-2-ol from propene.

steam, H_3PO_4 catalyst, 70 atm, 300°C

- (ii) State the reagent and conditions for the formation of propane-1,2-diol from propene.

cold, dilute, alkaline KMnO_4

- (iii) Show the structure of the product formed when propan-2-ol is heated under reflux with acidified potassium manganate(VII).



- (iv) Show the structure of the product formed when propane-1,2-diol is heated under reflux with acidified potassium manganate(VII) [4]

